2Q) A smart city project is collecting data from multiple sources, including:

-> Traffic sensors for vehicle counts and congestion levels.

-> IoT devices for monitoring air quality, noise and levels, and energy - consumption.

A) Smart City Project:

Star schema Design;

-> Fact table : SmartCity_Fact

-> Measures : Vehicle_count, Congestion level, Energy Consumption, Air quality, Noise level, Ridership, Punctually.

-> Dimensions : Time, location, Traffic sensor, IoT device, Transport System, Citizen Feedback.

-> Justification: Measuration stores numerical values for analysis, while dimensions provide different viewpoints for reporting.

<u>Measures:</u>

-> vehicle count

-> Congestion level

-> Energy Consumption

-> Air quality index

-> Noise level

-> Ridership, punctuality %.

<u>Dimensions:</u>

-> Time

-> location

-> Traffic sensor

-> IoT device

-> Transport mode

-> Citizen Feedback.

```
                    Time_Dim
                       |
Traffic_Dim ---- Fact_Smarcity ----Location_Dim
                       |
                  Transport_Dim
                       |
                   Citizen_Dim
```

3Q) Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and loan_Fact, sharing dimensions customer branch, time and account_type.

a) draw a galaxy schema diagram

b) If each dimensions has four levels, calculate.

A) a) Galaxy Schema

Fact Tables:

-> Transaction_Fact (transaction_count, amount)

-> Loan_Fact (loan_amount, intrest)

Shared dimensions:

-> Customer

-> Branch

-> Time

-> Account_type.

b) OLAP Operations:

=> Roll-up Time (year) and Branch, then calculate avg transaction amount loan amount.

c) Total cuboids:

Formula = $4^4 = 256$

Ans: (256 cuboids)

Fact Table:

-> vehicle count

-> Congestion level

-> Energy Consumption

-> Ridership

-> Punctually

3. Banking — Galaxy Schema:

Customer

|

Branch ------ Transaction_Fact ---- Time

|

Account_Type

Branch ------ Loan-Fact ------- Time

|

customer

|

Account-type.

Q) Manufacturing Company monitoring products quality and production efficiency. The data warehousing - dimensions;

Product : Category, type and batch

Factory : location, department and line number

Time : Year, quarter, month and week

Supplier : Region, rating, and contract type.

A) Roll-up:

→ Summarize defect rates by product category for the current yrs across all factories.

Drill-down:

→ Analyze defects rates by department and product Type for selected factory.

Slice:

→ Selects a suppliers with rating 4 or above and view production units.

Dice:

→ Selects specific product categories and factory locations to analyze defect and rate and supplier reliability.

Manufacturing - OLAP [operations]

Rollup:

Batch
↓
Type
↓
Category

drill-down:

Factory
↓
Department
↓
line number

Slice:

Supplier - Ratings ≥ 4
↓
Production units

Dice:

Selected Categories
↓
Selected locations
↓
Defect Rate
↓
Supplier reliability.

5Q) A retail chain manages inventory and sales data across multiple stores. The data warehouse contains :

dimensions:

Product: Category , sub category , and SKU

Store : Region, city and storeID

Time: Year, quarter, month and day

Customer: membership tier and agegroup and gender.

Measures:

→ Inventory turnover.

→ Total Sales revenue.

→ units sold.

A) Retail Chain:

Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.

Drill-down: Analyze electronics sales by subcategory and SKU in a selected region.

Slice: view sales revenue for male customers with Bronze membership across all stores

Dice: Analyze sales revenue and units sold for selected cities, product categories , and time periods.

Retail Chain - OLAP Operations:

Roll-up:

SKU → Subcategory → Category.

Drill-down:

Electronics
↓
Sub Category
↓
SKU

Slice:

Male
↓
Bronze members
↓
Sales Revenue

Dice:

Selected Cities
+
Selected Categories
+
Selected Time
↓
(Sales revenue -
units sold)

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
Student Name:	V. Dharanireddy	Reg. No.:	192311215
Section / Batch:	CSE -2023-2027	Date:	15-07-2026.

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
	GRAND TOTAL:		100 Marks

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?

(c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.

(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.

(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
  - Product: Category, type, and batch.
  - Factory: Location, department, and line number.
  - Time: Year, quarter, month, and week.
  - Supplier: Region, rating, and contract type.
- Measures:
  - Defect rate.
  - Production units.
  - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
  - Product: Category, subcategory, and SKU.
  - Store: Region, city, and store ID.
  - Time: Year, quarter, month, and day.
  - Customer: Membership tier, age group, and gender.
- Measures:
  - Inventory turnover.
  - Total sales revenue.
  - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

Code of Conduct

I ______________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks							
					Overall Total (100)		

Faculty In-charge	Course Coordinator	Head of Department
Signature: ____________	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________